

BUSINESS INSIGHTS REPORT

Amazon product analysis

Executive summary

The analysis covers 1,351 unique products after repeated product IDs are removed. It examines product categories, discount levels, customer ratings, and estimated commercial value. Because the source does not contain verified transactions, units sold, or geographic information, all financial figures should be interpreted as estimates and regional analysis is not available.

Metric	Result	Interpretation
Estimated revenue	Rs. 68.96bn	Value estimate derived from the cleaned product dataset
Estimated return loss	Rs. 4.26bn	Estimated returns equal 6.18% of estimated revenue
Estimated net revenue	Rs. 64.70bn	Estimated revenue after estimated return loss
Rating count	23.81M	Customer-review signal; not verified units sold
Average rating	4.09 / 5	Overall customer sentiment is positive

Priority insights and recommended actions

1. Concentration in Electronics

Electronics is the dominant estimated-revenue category, so its availability, product quality, and promotional performance have a disproportionate effect on the overall result.

Recommended action: Track category contribution and grow promising secondary categories to reduce reliance on one segment.

2. Ratings remain broadly positive across discount bands

Average ratings cluster around four stars across the dashboard heatmap. Discount depth alone therefore does not appear to be the primary driver of customer satisfaction.

Recommended action: Protect product quality and test discount changes selectively rather than assuming deeper discounts improve satisfaction.

3. Returns are a material estimated loss

Estimated return loss represents 6.18% of estimated revenue. The return rate is modelled from rating information, so this is directional rather than an audited loss figure.

Recommended action: Review low-rated products, strengthen product descriptions, and prioritise seller-quality checks before expanding promotions.

4. A small group of products leads the ranking

The Top 10 chart identifies products with unusually high estimated value. Their performance has an outsized effect on category totals.

Recommended action: Use the ranking to prioritise listing quality, stock planning, and price monitoring.

Dashboard and analytical coverage

Project requirement	How it is addressed
Data cleaning and preprocessing	Numeric price, discount, rating, and rating-count handling; product-level deduplication; discount-band grouping.
Sales KPI analysis	Estimated revenue, return loss, net revenue, return rate, rating count, and average rating cards.
Interactive charts	Category and discount-band filters update charts and KPI cards.
Product performance	Category comparison, rating heatmap, and Top 10 products by estimated revenue.
Regional sales analysis	Not available: the source dataset has no valid geographic field. Category segmentation is used instead.
Business insights	Action-oriented findings cover category concentration, discount strategy, quality, returns, and priority products.

Methodology and limitations

Metric interpretation. The cleaned dataset includes estimated-revenue and estimated-return fields. These are not audited sales figures. Rating count is used as a customer-interest indicator, not as a substitute for unit sales.

Product-level reporting. Repeated product IDs are removed in the web dashboard before KPI aggregation, preventing duplicate products from inflating totals.

Regional limitation. No region, state, city, customer address, or geographic identifier exists in the supplied dataset. A map or regional sales claim would require additional data and is intentionally excluded.

Recommended next data collection. Add transaction date, verified units sold, order value, return reason, seller, inventory availability, and customer geography to strengthen future analysis.

Conclusion

The dashboard is useful for exploratory category and product analysis. Its strongest value is identifying high-impact categories and products while keeping the limitations of estimated metrics explicit. Verified sales and geographic data should be added before using the findings for operational or financial decisions.

Source: amazon_cleaned.csv supplied with the project.